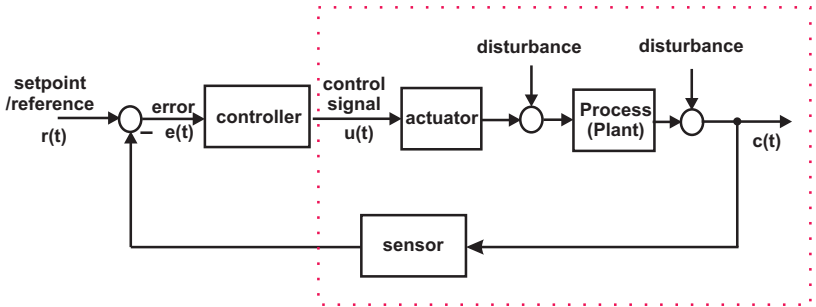
The Design of Feedback Control Systems

Paula Raica
Department of Automation
Dorobantilor 71-73, room C21,
email: Paula.Raica@aut.utcluj.ro

Alexandru Codrean
Department of Automation
Dorobantilor 71-73, room C23,
email: alexandru.codrean@aut.utcluj.ro

Technical University of Cluj-Napoca

Control system - fundamental structure



Introduction

- Main goals of a control system: **stability**, reference/setpoint **tracking**, disturbance rejection.
- The performance of a feedback control system: stability, accuracy in tracking the reference input, less sensitive to system parameter changes, reduce the effect of undesirable disturbances.
- The design of a control system = the arrangement of the system structure and the selection of suitable components and parameters.
- To alter the system response, an additional component is inserted within the structure of the feedback system: *controller* or *compensator*.

Approaches to Control System Design

- Performance specifications for a control system:
  - In time-domain: overshoot, settling time, steady state error.
  - In frequency domain: resonant frequency, the bandwidth, the phase margin.
- Assume that the process has been improved as much as possible and the transfer function representing the process is unalterable.
- *A controller is actually an algorithm* implemented and running on a digital device, an analog electronic circuit, or even mechanical circuits.
- Sometimes the physical device on which the control algorithm runs is also called a controller (ex. PLC - programmable logical controller)

Approaches to Control System Design

Control design - how hard can it be?



Theory vs. practice gap....

Approaches to Control System Design

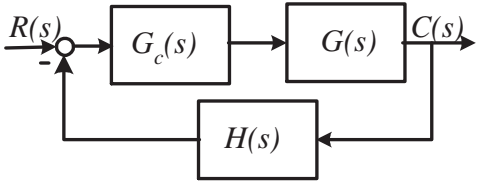
- Depending on the type of system that we want to control, there are a multitude of controller types and design methods (see *Encyclopedia of Systems and Control, Editors John Baillieul & Tariq Samad, Springer, 2021,* for a comprehensive look with applications)
- In this course we will focus on the two most commons controllers used in applications: **lead-lag compensators** and **PID**.
- As design approaches, a general approach is the root locus method, but we will discuss also about steady state error based design, and also tuning rules.

Root-locus approach

- RL: a graphical method for determining the locations of all closed-loop poles as some parameter (usually the controller gain) is varied from zero to infinity.
- LR yields a clear indication of the effects of parameter adjustment and information on the transient response.
- For control system design purposes it is often necessary to reshape the RL to meet the performance specifications.
- In the design via the RL method: the root loci of the system are reshaped through the use of a compensator so that a pair of dominant closed-loop poles can be placed at the desired location.

Cascade compensation networks

The compensator $G_c(s)$ is cascaded with the unalterable process $G(s)$ in order to provide a suitable loop transfer function $G_c(s)G(s)H(s)$.

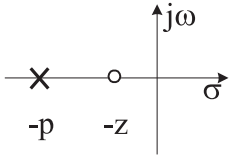


$$G_c(s) = \frac{k\prod_{i=1}^{M}(s+z_i)}{\prod_{j=1}^{N}(s+p_j)}$$

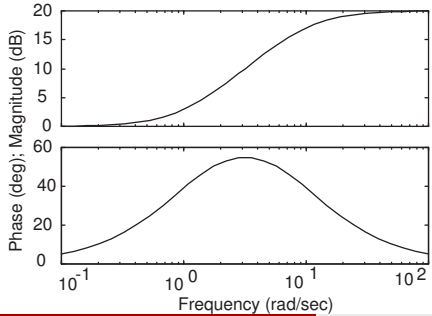
- The compensator $G_c(s)$ can be chosen to alter the shape of the root locus or the frequency response.
- The problem reduces to the selection of the poles and zeros of the compensator.
- Consider a first-order compensator $G_c(s)$.

Phase-lead compensator

$$G_c(s) = \frac{k(s+z)}{s+p}, \qquad |z| < |p|$$



The frequency response is:



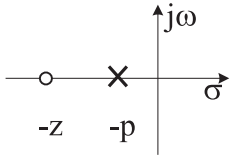
May be used to alter the transient state of the closed-loop system

Example (lead element)

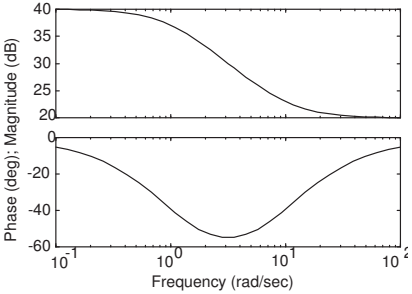
$$G(s) = \frac{s+1}{s+10}$$

Phase-lag compensator

$$G_c(s) = \frac{k(s+z)}{s+p}, \qquad |z| > |p|$$



The frequency response:



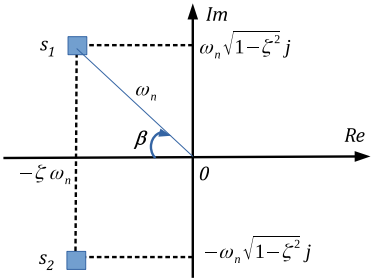
May be used to alter the steady-state error of the closed-loop system

Example (lag element)

$$G(s) = \frac{s+10}{s+1}$$

Reminder: poles and transient response characteristics

- The poles of a (2nd order) system can be related to the characteristics of the transient response (settling time, overshoot)



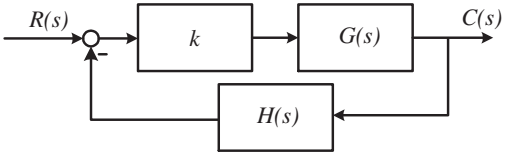
$$t_s = \frac{4}{\zeta\omega_n}$$

$$M_P = e^{-\pi\zeta/\sqrt{1-\zeta^2}}, \quad \cos\beta = \zeta$$

$$t_r = \frac{\pi-\beta}{\omega_d}, \quad t_p = \frac{\pi}{\omega_d}$$

$$(\omega_d = \omega_n\sqrt{1-\zeta^2})$$

Reminder: root locus



- Open-loop transfer function: $H_d(s) = kG(s)H(s)$
- Closed-loop transfer function: $\dfrac{C(s)}{R(s)} = \dfrac{kG(s)}{1 + kG(s)H(s)}$
- The root locus: a plot of the **closed-loop poles** for $k \in [0, \infty)$, i.e. the roots of: $1 + kG(s)H(s) = 0$.
- **Angle condition:** $\angle kG(s)H(s)|_{s \in LR} = -180^o$
- **Magnitude condition**: $|kG(s)H(s)|_{s \in LR} = 1$

Phase-lead compensation based on root locus

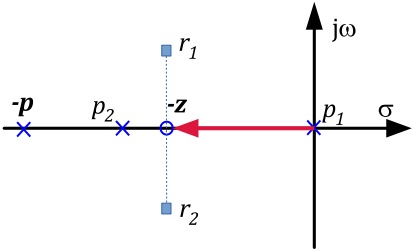
- Time-domain specifications:
  - location of closed-loop poles,
  - overshoot,
  - settling time,
  - damping factor
  - etc.
- Consider a design problem in which the original system either is unstable for all values of gain or it is stable but has undesirable transient response characteristics
- Insert an appropriate lead compensator in cascade with the feed-forward transfer function.

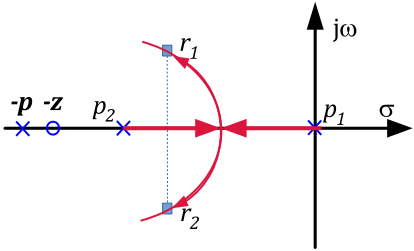
Phase-lead compensation based on root locus

1. Closed-loop specifications $\rightarrow$ desired location for the dominant roots ($r_{1,2}$).
2. Sketch the uncompensated root locus and determine whether the desired root locations can be realized with an uncompensated system
3. Select a compensator $G_c(s) = \frac{k(s+z)}{s+p}$, $p > z > 0$.
4. Place in the complex plane all the known poles and zeros of the open-loop system.
5. Place the zero of the phase-lead compensator directly below the desired root location (or to the left of the first two real poles)
6. Determine the pole location so that the total angle at the desired root location is -180° (the angle condition is fulfilled for $r_{1,2}$, i.e. the desired poles are on the RL of the compensated system).
7. Evaluate the total system gain from the magnitude condition for $s = r_{1,2}$.

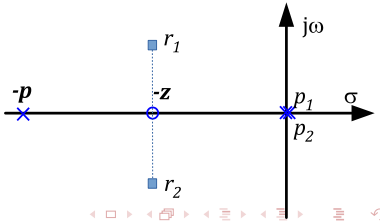
Phase-lead compensation

Add the compensator zero to the left of the first two real poles (so $r_{1,2}$ are the dominant poles).





If the compensator zero is between (p_1 and p_2), the RL branch (red) is closer to the origin than $r_{1,2}$, $\Rightarrow$ the pole on this branch will be dominant.



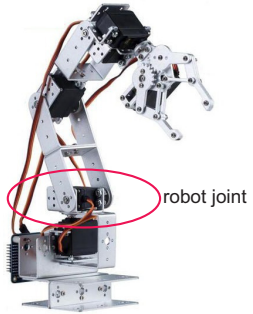
Phase-lead compensation

- If $G(s)$ is the process transfer function (known) and the compensator is
  $G_c(s) = \dfrac{k(s+z)}{s+p}$, with k and $p > z > 0$ unknown
  - Choose the zero of the compensator $-z$.
  - Compute the pole of the compensator $-p$ from the angle condition
  - Compute the gain k from the magnitude condition
- If $G(s)$ is the process transfer function (known) and the compensator is partially known (the pole or the zero is known/predefined):
  - Compute the pole **or** the zero of the compensator from the angle condition
  - Compute the gain k from the magnitude condition.

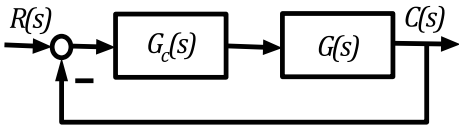
Phase-lead compensation: example 1 - robot control

Let us consider a motion control problem, where a robot joint is actuated by a DC motor. The input is the torque u and the output is the angular position c. The transfer function is:

$$G(s) = \frac{c(s)}{u(s)} = \frac{1}{s^2}$$



The system $G(s)$ is unstable (two poles in 0).
We adopt the feedback control structure:



We will further design a closed-loop control system with the specifications:

- Settling time $t_s \leq 4$ seconds
- Percent overshoot for a step input $M_p \leq 35\%$.

Phase-lead compensation: example 1

Consider a simple P compensator $G_c(s) = k$ and try to obtain the value of the gain k so that the design specifications are met.

- The closed-loop transfer function with the P-compensator is:

$$T(s) = \frac{kG(s)}{1 + kG(s)} = \frac{k\frac{1}{s^2}}{1 + k\frac{1}{s^2}} = \frac{k}{s^2 + k}$$

  $\Rightarrow$ undamped oscillations.
- Select a lead compensator $G_c(s)$, where

$$G_c(s) = \frac{k(s + z)}{s + p}, \qquad p > z > 0$$

- Use the desired specifications for the closed-loop system to obtain a pair of dominant poles.

Phase-lead compensation: example 1

The admissible area for the dominant closed-loop poles:

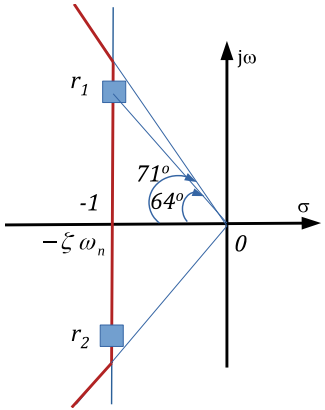
- The damping ratio of the compensated system ζ:

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \leq 0.35 \quad \Rightarrow \zeta \geq 0.325$$

$$\Rightarrow \beta \leq \arccos\zeta = 71^o$$

- The settling time requirement is

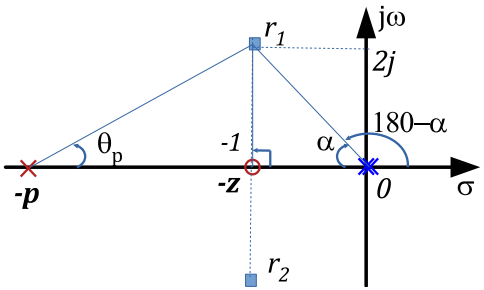
$$t_s = \frac{4}{\zeta\omega_n} \leq 4 \quad \Rightarrow \zeta\omega_n \geq 1 \quad \Rightarrow -\zeta\omega_n \leq -1$$



- All complex poles located inside an angle of 71° and to the left of the vertical line at -1 will fulfill the specifications.
- $r_{1,2} = -1 \pm 2j$, where the angle is $\alpha \approx 64^o$.

Phase-lead compensation: example 1

- Place all the (known) open-loop poles and zeros in the complex plane. Place the dominant poles $r_{1,2}$.



- Place the compensator zero below r_1 (it is on the left of two open-loop poles (0, 0)): $-z = -1$.
- Place the compensator pole on the negative real axis and determine its value so that the dominant poles $r_{1,2}$ are on the RL of the compensated system, i.e. the angle condition is satisfied for $r_{1,2}$:

Phase-lead compensation: example 1

- Angle condition: $\boxed{\angle G_c(s)G(s)|_{s=r_1} = -180^0}$.

$$\angle \frac{k(s+z)}{s+p} \cdot \frac{1}{s^2}\bigg|_{s=r_1} = -180^0$$

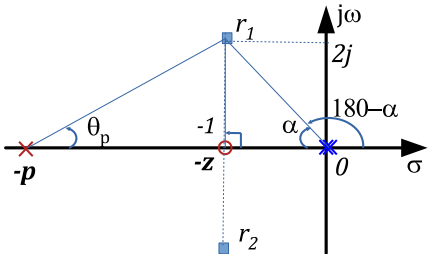
$$\angle G_c(s)G(s)|_{s=r_1} = \angle k + \angle(r_1+z) - \angle(r_1+p) - 2\angle(r_1) = -180^0$$

- From the figure: $\angle k = 0$, $\angle(r_1+z) = \angle(r_1+1) = 90^o$, $\angle(r_1+p) = \theta_p$, $\angle(r_1) = 180^o - \alpha$, $\alpha = 64^0$.

$$\angle G_c(s)G(s)|_{s=r_1} = 90^o - \theta_p - 2(180^o - 64^o) = -142^o - \theta_p = -180^o$$

$$\Rightarrow \theta_p = 38^o.$$

Phase-lead compensation: example 1



- In the right triangle $(r_1,\ -z,-p)$:

$$\tan\theta_p = \tan 38^o = \frac{2}{p-1} \quad \Rightarrow\ p-1 = \frac{2}{\tan 38^o} \quad \Rightarrow\ p = 3.55$$

- The compensator is now: $G_c(s) = \dfrac{k(s+1)}{s+3.55}$.
- If $k \in [0,\ \infty)$ the RL of the closed-loop system will pass through $r_{1,2}$.

Phase-lead compensation: example 1

- The open-loop transfer function of the compensated system is:

$$G_c(s)G(s) = \frac{k(s+1)}{s^2(s+3.55)}$$

- The gain k is evaluated from the magnitude condition:

$$\boxed{|G_c(s)G(s)|_{s=r_1} = 1} \qquad \Rightarrow \left|\frac{k(s+1)}{s^2(s+3.55)}\right|_{s=-1+2j} = 1$$

$$k = \left|\frac{s^2(s+3.55)}{s+1}\right|_{s=-1+2j} = \left|\frac{(-1+2j)^2(-1+2j+3.55)}{-1+2j+1}\right|$$

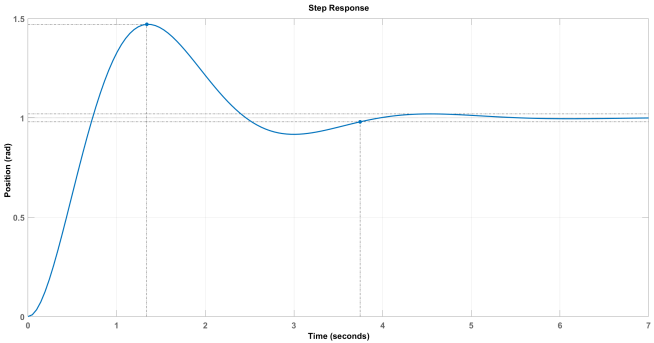
$$k = \frac{(\sqrt{5})^2 \cdot \sqrt{2.55^2+2^2}}{2} = 8.1$$

- Finally, we obtain the phase-lead compensator that meets the design requirements

$$G_c(s) = \frac{8.1(s+1)}{s+3.55}$$

Phase-lead compensation: example 1

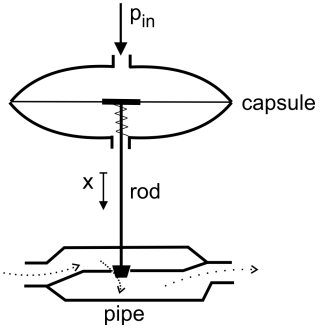
- Simulations results for a step reference input:



- The discrepancy between the desired overshoot and the one obtained is due to the extra zero introduced by the controller. Further tuning the controller gain and pole can decrease this overshoot.

PI compensator based on RL: example 2 - a pneumatic system

- The force due to the input pressure $p_{in}(t)$ pushes on the flexible membrane, which makes the rod move with the position $x(t)$. The shutter at the end of the rod can further influence the flow through a pipe.
- For a sufficiently fast air filling inside the capsule, we can consider that the air pressure inside the capsule is $p(t) = p_{in}(t)$.



PI compensator based on RL: example 2 - a pneumatic system

- The mechanical part is described by Newton's second law of dynamics:

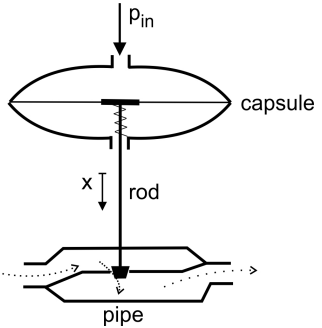
$$m\frac{d^2x(t)}{dt^2} = -b\frac{dx(t)}{dt} - kx(t) + Sp(t),$$

where m is the mass, b is the friction coefficient, k is the elastic coefficient, and S is the membrane area.

- For the parameter values $m = 1, b = 4.5, k = 2, S = 1$, the transfer function of the system is

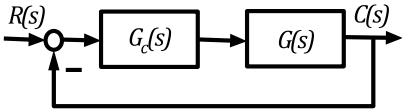
$$G(s) = \frac{x(s)}{p_{in}(s)} = \frac{1}{s^2 + 4.5s + 2}$$

.



PI compensator based on RL: example 2 - a pneumatic system

- Consider a closed-loop control system with the plant $G(s) = \dfrac{1}{(s+\frac{1}{2})(s+4)}$.
- Determine a PI controller $G_c(s) = K_P + \dfrac{K_I}{s}$, so that the dominant closed-loop poles are $r_{1,2} = -0.5 \pm j$.

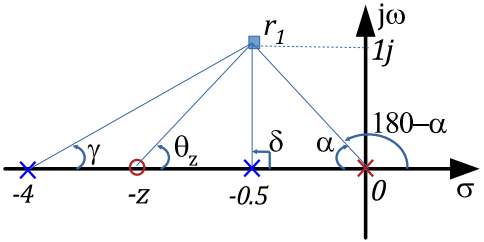


- The controller can be written (to emphasize the poles and zeros):

$$G_c(s) = K_P + \frac{K_I}{s} = K_P \frac{s + K_I/K_P}{s} = k\frac{s+z}{s}$$

- The compensator zero (z) will be calculated from the angle condition and the gain (k) from the magnitude condition.

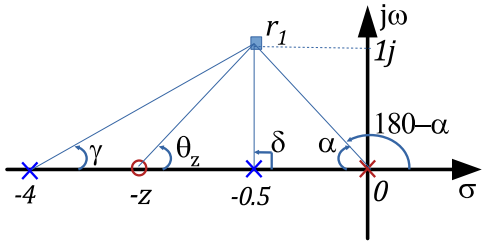
PI compensator based on RL: example 2



- Place the poles that are known: the poles of the process (-4 and -0.5) and the pole of the compensator (0).
- The zero will be *calculated* from the angle condition:

$$\angle G_c(s)G(s)|_{s=r_1} = -180^o \;\Rightarrow\; \angle \frac{k(s+z)}{s(s+\frac{1}{2})(s+4)}\bigg|_{s=r_1} = -180^o$$

PI compensator based on RL: example 2



$$\angle k + \angle(r_1 + z) - \angle r_1 - \angle(r_1 + \frac{1}{2}) - \angle(r_1 + 4) = -180^o$$

$$0 + \theta_z - (180^o - \alpha) - \delta - \gamma = -180^o$$

$$\theta_z - (180^o - \underbrace{\arctan\frac{1}{0.5}}_{63.4^o}) - 90^o - \underbrace{\arctan\frac{1}{4-0.5}}_{15.9^o} = -180^o$$

$$\theta_z = 42.5^o \quad \Rightarrow \quad \Delta(r_1,\ -0.5,\ -z): \quad \tan\theta_z = \tan 42.5 = \frac{1}{z - 0.5} \Rightarrow z = 1.59$$

PI compensator based on RL: example 2

- The compensator is now: $G_c(s) = \dfrac{k(s+1.59)}{s}$.
- The gain k is calculated from the magnitude condition:

$$|G_c(s)G(s)|_{s=r_1} = \left| \frac{k(s+1.59)}{s(s+\frac{1}{2})(s+4)} \right|_{s=-0.5+j} = 1$$

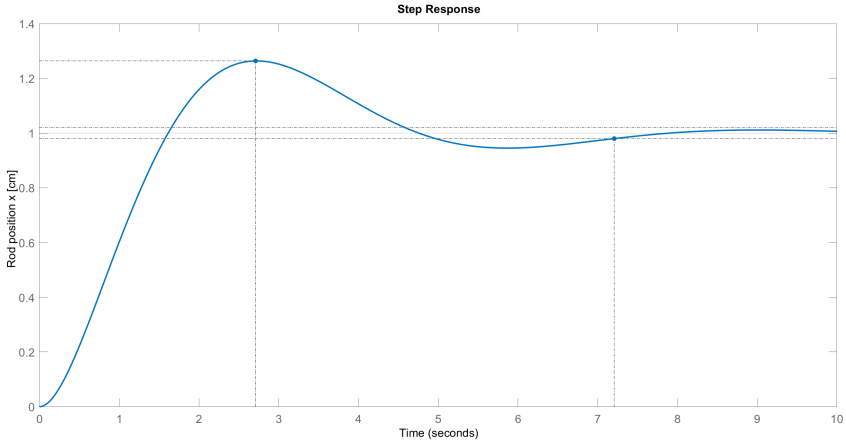
$$k = \left| \frac{(-0.5+j)(-0.5+j+\frac{1}{2})(-0.5+j+4)}{(-0.5+j+1.59)} \right|$$

$$k = \frac{\sqrt{0.5^2+1^2} \cdot 1 \cdot \sqrt{3.5^2+1^2}}{\sqrt{1.09^2+1^2}} = 2.75$$

- The compensator: $G_c(s) = 2.75\dfrac{s+1.59}{s} = 2.75 + \dfrac{4.37}{s}$

PI compensator based on RL: example 2 - a pneumatic system

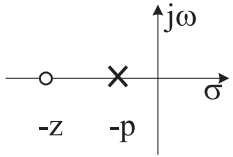
- Simulations results for a step reference input:



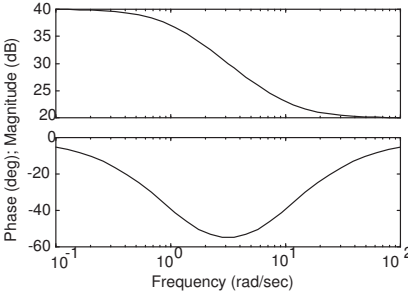
Phase-lag compensation using root locus

Phase-lag compensator

$$G_c(s) = \frac{k(s+z)}{s+p}, \qquad |z| > |p|$$



The frequency response:



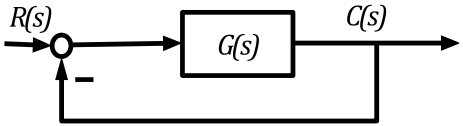
May be used to alter the steady-state error of the closed-loop system

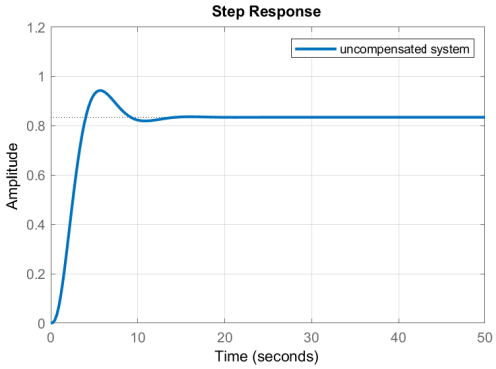
Example (lag element)

$$G_c(s) = \frac{s+10}{s+1}$$

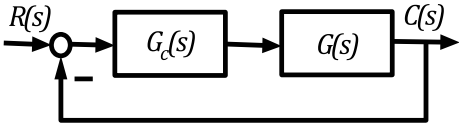
Compensating the steady-state error

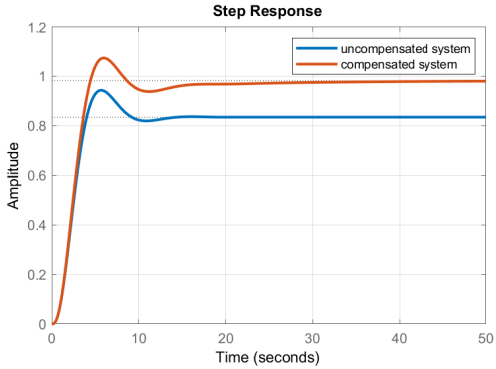
Uncompensated system





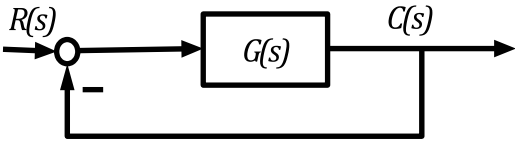
Compensated system





Steady-state error

- (see Chapter 3, section 3.5.3 on static error constants)
- Consider a closed-loop negative feedback control system



$$G(s) = \frac{k\prod_{i=1}^{M}(s+z_i)}{s^N\prod_{i=1}^{Q}(s+p_i)}$$

  - $G(s)$ = the open-loop transfer function
  - unity feedback
  - N = type number
- The steady-state error:

$$e_{ss} = \lim_{t\to\infty} e(t) = \lim_{t\to\infty}(r(t) - c(t))$$

$$e_{ss} = \lim_{s\to 0} s(R(s) - C(s)) = \lim_{s\to 0} sR(s)(1 - \frac{G(s)}{1+G(s)}) = \lim_{s\to 0}\frac{sR(s)}{1+G(s)}$$

Steady-state error

Step input: $r(t) = 1 \Rightarrow R(s) = \dfrac{1}{s}$

$$e_{ss} = \lim_{s \to 0} s \frac{1}{s} \frac{1}{1 + G(s)} = \frac{1}{1 + G(0)} = \frac{1}{1 + K_p}$$

$$K_p = \lim_{s \to 0} G(s) \ = \text{the position error constant}$$

- $N = 0$:

$$G(s) = \frac{k \prod_{i=1}^{M} (s + z_i)}{\prod_{i=1}^{Q} (s + p_i)} \quad \Rightarrow \ K_p = const \Rightarrow \ e_{ss} = const$$

- $N \geq 1$:

$$G(s) = \frac{k \prod_{i=1}^{M} (s + z_i)}{s^N \prod_{i=1}^{Q} (s + p_i)} \quad \Rightarrow \ K_p = \infty \Rightarrow \ e_{ss} = 0$$

Steady-state error

Ramp input: $r(t) = t \Rightarrow R(s) = \dfrac{1}{s^2}$

$$e_{ss} = \lim_{s\to 0} s\frac{1}{s^2}\frac{1}{1+G(s)} = \lim_{s\to 0}\frac{1}{s(1+G(s))} = \lim_{s\to 0}\frac{1}{sG(s)} = \frac{1}{K_v}$$

$$K_v = \lim_{s\to 0} sG(s) \ = \text{the velocity error constant}$$

- $N = 0$: $G(s) = \dfrac{k\prod_{i=1}^{M}(s+z_i)}{\prod_{i=1}^{Q}(s+p_i)} \quad \Rightarrow \ K_v = 0 \ \Rightarrow \ e_{ss} = \infty$
- $N = 1$:

$$G(s) = \frac{k\prod_{i=1}^{M}(s+z_i)}{s\prod_{i=1}^{Q}(s+p_i)} \quad \Rightarrow \ K_v = const \Rightarrow \ e_{ss} = const$$

- $N \geq 2$:

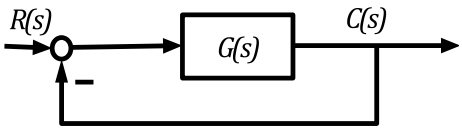
$$G(s) = \frac{k\prod_{i=1}^{M}(s+z_i)}{s^N\prod_{i=1}^{Q}(s+p_i)} \quad \Rightarrow \ K_v = \infty \Rightarrow \ e_{ss} = 0$$

Phase-lag comp. based on RL

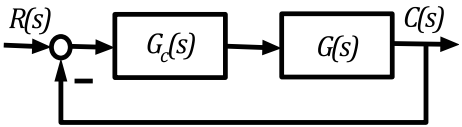
- Consider the problem of finding a suitable compensator for the case where the system exhibits satisfactory transient-response characteristics but unsatisfactory steady-state characteristics
- Compensation in this case essentially consists of increasing the open-loop gain without appreciably changing the transient response characteristics.
- To avoid an appreciable change in the root loci the angle contribution of the lag element should be limited to a small amount, say 5°;
- To assure this, we place the pole and zero of the lag network relatively close together and near the origin of the s-plane.

Compensating the steady-state error

Uncompensated system (before lag compensation)



Compensated system (after lag compensation)



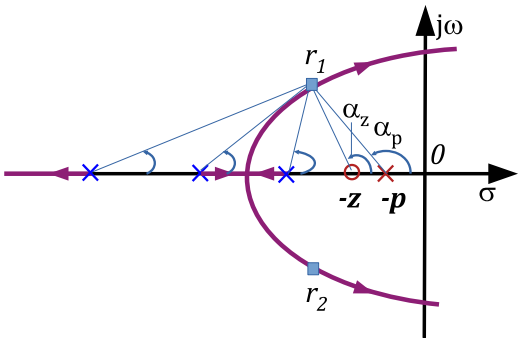
- Add a lag compensator $G_c(s) = \dfrac{s+z}{s+p}$, where $z > p > 0$
- $r_{1,2}$ are the dominant poles ($\Rightarrow$ the transient response) of the closed-loop system before lag compensation.
- The lag compensator should not change the transient response (keep the dominant poles in approximately the same position)
- $r_{1,2}$ satisfy the angle condition for the uncompensated RL:

$$\angle G(s)|_{s=r_1} = -180^o$$

Phase-lag comp. based on RL

- The angle condition for the compensated RL:

$$\angle G(s)G_c(s)|_{s=r_1^*} = \angle G(s)\frac{s+z}{s+p}|_{s=r_1^*} = \angle G(s) + \underbrace{\angle s+z - \angle s+p}_{<5^o}|_{s=r_1^*} = -180^o$$



- $-z$ and $-p$ should be very close to each other to keep the dominant poles in almost the same location.

Phase-lag comp. based on RL

Consider a type 0 system for a step input, or a type 1 system for a ramp input.

- For the uncompensated system:

$$K_p = \lim_{s\to 0} G(s), \quad e_{ss} = \frac{1}{1+K_p}$$

$$K_v = \lim_{s\to 0} sG(s), \quad e_{ss} = \frac{1}{K_v}$$

- For the compensated system:

$$K_{pcomp} = \lim_{s\to 0} G(s)\frac{s+z}{s+p} = \frac{z}{p}K_p, \quad e_{ss} = \frac{1}{1+K_{pcomp}}$$

$$K_{vcomp} = \lim_{s\to 0} sG(s)\frac{s+z}{s+p} = \frac{z}{p}K_v, \quad e_{ss} = \frac{1}{K_{vcomp}}$$

- The ratio $\frac{z}{p}$ should be large to improve the steady-state error.

Phase-lag comp. based on RL

- The zero $(-z)$ and the pole $(-p)$ of the compensator should be close to each other to maintain location of the dominant closed-loop poles (and the transient response characteristics)
- The zero $(-z)$ and the pole $(-p)$ of the compensator should have a large ratio $\frac{z}{p}$ to improve the steady-state error.
- Solution: place them close to the origin *comparing to the location of the dominant poles*:
  - Choose the zero of the compensator small comparing to the natural frequency of the dominant poles: $z \approx \frac{\omega_n}{10}$
  - Compute the pole from the ratio $\frac{z}{p}$.

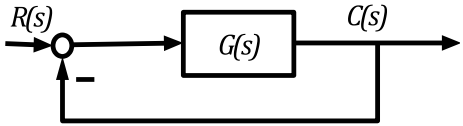
The algorithm

1. Determine the transient performance specifications for the system and locate suitable dominant root locations on the uncompensated root locus that will satisfy the specifications
2. Determine the dominant poles of the uncompensated system $r_{1,2}$
3. Calculate the system error constant at the dominant root location for the uncompensated system (before lag compensation): K_p or K_v.
4. Calculate the ratio z/p from the desired error constant and the uncompensated error constant.
5. Locate the zero close to the origin with respect to ω_n of dominant poles (for example $z \approx \frac{\omega_n}{10}$)
6. Determine the pole $-p$ from the ratio z/p.

Example

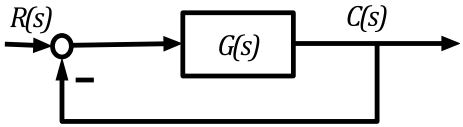
- Consider the line-following problem for a two wheeled Segway robot. Because a constant velocity is usually preferred, we adopt a **ramp** reference input ($r(t) = t,\ t \geq 0$).
- The open loop transfer function and the closed-loop system are:

$$G(s) = \frac{5}{s(s+2)}$$



- Design a lag compensator so that the velocity error constant is $K_{vcomp} = 20$ (or the steady-state error for a ramp input is $e_{ss} = \frac{1}{20} = 0.05$).

Example



- Before the lag compensation (for the uncompensated system), the closed-loop transfer function and the closed-loop poles:

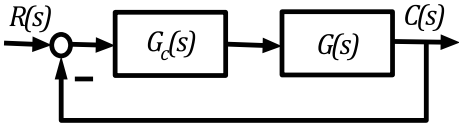
$$G_0(s) = \frac{G(s)}{1 + G(s)} = \frac{5}{s^2 + 2s + 5}, \qquad r_{1,2} = -1 \pm 2j$$

- The static velocity error constant is:

$$K_v = \lim_{s \to 0} s \frac{5}{s(s+2)} = \frac{5}{2} = 2.5 \qquad (e_{ss} = \frac{1}{2.5} = 0.4)$$

Example

- Add a lag compensator:



$$G_c(s) = \frac{s+z}{s+p} \text{ with } z > p > 0$$

- The velocity error constant for the compensated system:

$$K_{vcomp} = \lim_{s\to 0} s\frac{5}{s(s+2)}\frac{s+z}{s+p} = 2.5\frac{z}{p} = 20$$

- The ratio $\frac{z}{p} = \frac{20}{2.5} = 8$.
- The natural frequency of the dominant poles $r_{1,2}$: $\omega_n = \sqrt{1^2+2^2} = \sqrt{5}$.
- Choose the zero so that $z < \omega_n/10$, for example: $z = 0.1$ (the zero at $-z = -0.1$) and the pole is:

$$\frac{z}{p} = 8 \Rightarrow p = \frac{0.1}{8} = 0.0125 \Rightarrow -p = -0.0125 \Rightarrow G_c(s) = \frac{s+0.1}{s+0.0125}$$

Example

The difference of the angles from $-p$ and $-z$ at the desired root is approximately 1°, and therefore $s = -1 \pm 2j$ is still the location of the dominant roots.

